A Project Report

On

VEHICLE DOCUMENT VALIDATION THROUGH AUTOMATED NUMBER PLATE RECOGNITION

Submitted by

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CERTIFICATE

We **Revanth D Rampal (R180753)**, **P Divya (R180897)** here declare that this report entitled "**VEHICLE DOCUMENT VALIDATION THROUGH AUTOMATED NUMBER PLATE RECOGNITION**" submitted by us under guidance and supervision of **Mr. M Muni Babu is** a bonafied work. I also declare that it has not been submitted previously in part or in full to this university or institution for the award of any degree.

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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose constant guidance and encouragement crown all the efforts success.

I am extremely grateful to our respected Director, Prof. K. SANDHYA RANI for fostering an excellent academic climate in our institution.

I also express my sincere gratitude to our respected Head of the Department Mr. N. SATHYANANDARAM for his encouragement, overall guidance in viewing this project a good asset and effort in bringing out this project.

I would like to convey thanks to our guide at college Mr. M. MUNI BABU for his guidance, encouragement, co-operation and kindness during the entire duration of the course and academics.

My sincere thanks to all the members who helped me directly and indirectly in the completion of project work. I express my profound gratitude to all our family members for their encouragement.

ABSTRACT

Automatic Number Plate Recognition (ANPR) is designed for automatic authorization of vehicles by vehicle number plate. The objective is to design an automatic vehicle document authorization system by using a number plate. The system first detects the moving vehicle using You Only Look Once (YOLO). Then it captures the vehicle image in different frames. The obtained vehicle image is then converted to gray scale the captured image using Region of Interest captures only the number plate region. Using different algorithms the number plate area is filtered. The filtered image is segmented character by character and is stored in different variables. The final stage of ANPR is Optical Character Recognition (OCR), the output data is stored. The output data is compared with the records on a database present which cross-checks for the documents are valid or expired. The graphical representation shows the overall details of the vehicles passed. It is observed the system detects the number plate and successfully recognizes the number plate.

Keywords: Automatic Number Plate Recognition (ANPR), Optical Character Recognition (OCR), License Plate.

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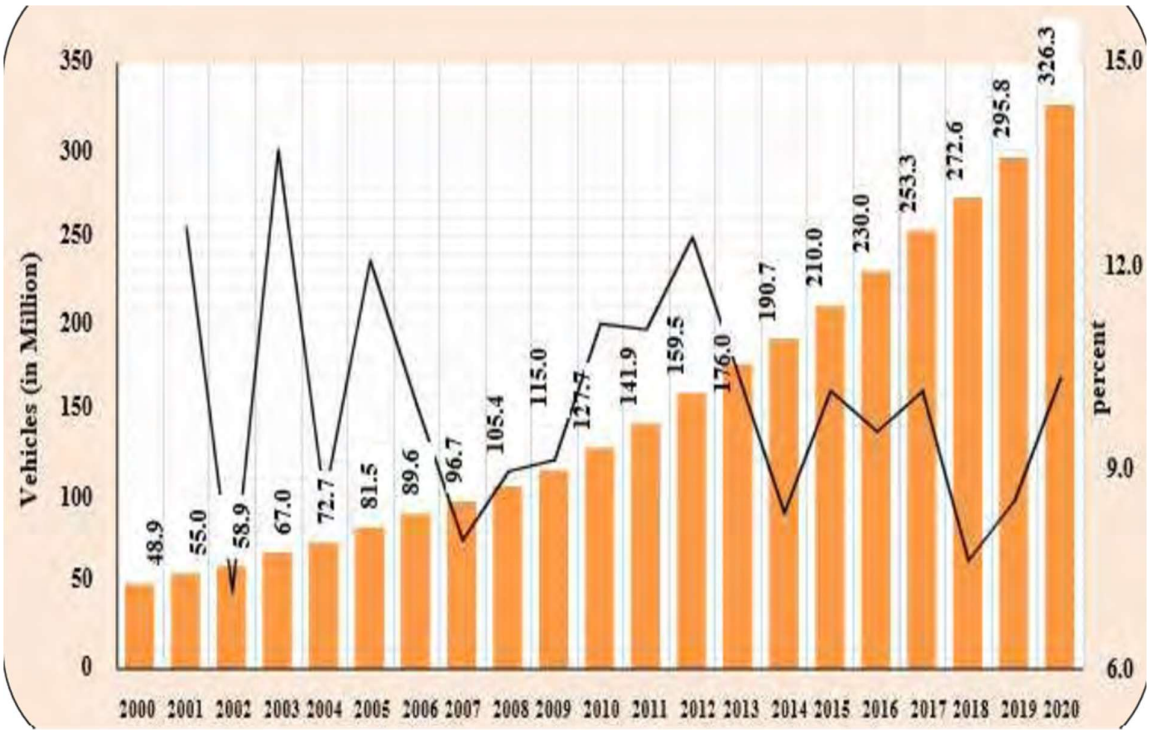
I. INTRODUCTION

The Automatic Number Plate Recognition (ANPR) was first invented in the year 1976 by Police Scientific Development Branch (PSDB), United Kingdom [3]. The growth of the ANPR gained with the improvement of the digital camera. ANPR can be used in many areas such as parking management, traffic control, recovery of stolen vehicles it can also be used to locate unauthorized vehicles, tracking any suspicious vehicles etc. ANPR uses different algorithms like edge detection and Hough transformation which helps in the extraction of data form the number plate. The ANPR system works in four steps, step one detection and capturing of the vehicle, the second step detection and extraction of number plate (Region of Interest), the third step using image segmentation technique segment to individual character and the final step Optical Character Recognition (OCR) to recognise the individual segment and store it in different variables. The image processing techniques uses edge extraction and Hough transformation but the recent advanced protocol involves implementation of the Neural Networks [3].

Automatic Number Plate Recognition systems are a proven solution for various security Forces and administrative authorities around the world. In India the growth of vehicles increases every year which leads to many violation of traffic rules. The graph represents the growth of vehicles per year quoted from the “Road Year Transport Book” Ministry of Road Transport and Highways [4]. The graph shows number of vehicles owned every year.



Graph 1.1: Total number of Registered Vehicle (in Million) in India [4]

Problem Statement:

The objective of the project is to create and implement an Automatic Number Plate Recognition system (ANPR) that can effectively and accurately track vehicles. The ANPR system with the aid of image processing and Optical Character Recognition (OCR) using different algorithm and vehicle database makes it possible to monitor vehicles and traffic at a large scale. The ANPR allows the traffic surveillance to deal with several problems like controlling speed, monitoring traffic, criminal investigation etc. The purpose of ANPR is to track details of the vehicles in real time using number plate.

Purpose:

The main purpose of the project is to reduce human effort to track and verify the vehicle documents. The system also reduces the potential of manual errors caused by human. ANPR technology increases Efficiency & saves time to process huge data. By using ANPR we benefit from increased accuracy, time savings, and reduction in fraud.

Scope:

Automatic Number Plate Recognition (ANPR) has a broad scope that includes various aspects across different sectors. The ANPR technology involves capturing number plate and processing it to extract the character on the plate. The key areas where the ANPR can be utilized:

- Law enforcement
- Parking Management
- Security and Surveillance
- Traffic monitoring and Management
- Toll collection
- Recovery of stolen vehicles

II. LITERATURE REVIEW

Automatic Number Plate Recognition was present from the year 1976 the development in this scope has grown in various aspects. The algorithm for the extraction of data from the number plate have developed good accuracy over the time of research. The ANPR technology has gained attention due to its application in different domains such as the parking system, traffic management, vehicle tracking system. The review aims to provide an overview of the research areas and advancement.

By comparing the previous work done in the field of the Automatic Number Plate Recognition we come to know that different algorithms and methodology was used to achieve different levels of accuracy some of the works are stated below with the algorithm used to achieve accuracy [2].

Saqib Rasheed [2012] proposed "Automated number plate recognition using though lines and template matching" using Hough Transformation and Template matching model which gave accuracy of 90.62 percent.

Alice N Cheeran [2017] proposed "Automatic Car Number Plate Recognition" using Ant Colony Optimization Algorithm which gave 94 percent accuracy.

Sumanta Subhadhira [2014] proposed "License Plate Recognition Application using Extreme Learning Machines" using extreme learning Method which is used for segmentation which offered 89.05 percent accuracy.

Yoshihiro Shima [2016] submitted "Extraction of Number Plate Images Based on Image Category Classification using Deep Learning" which used extraction techniques combine with morphological image processing and deep learning. This system offered 89.7 percent of accuracy.

Wei-chen Liu [2017] proposed "A Hierarchical Plate Recognition System using Supervised K-means and Support Vector Machine" supervised k-means and SVM to recognized blurred and tilted licence plate which province by 98.89 percentage.

Yongsheng Li [2019] proposed "Vehicle License Plate Recognition Combining MSER and Support Vector Machine in a Complex Environment" which is based on MSER and SVM algorithm which is reported to give above 90 percent of accuracy.

Yang Guang [2011] proposed "License Plate Character Recognition Based on Wavelet Kernel LS-SVM" which used wavelet LS-SVM (least square svm) giving accuracy of 98.3 percent.

Dinesh Bhardwaj [2014] proposed "Comparison of ML Algorithms for Identification of Automated Number Plate Recognition" which is the new methodology which includes six parts Random forest classifiers, Rep tree, IBK, K-star, universal discourser which gave an accuracy of 99 percent and is the best performing algorithm.

Mahima Satsangi [2018] proposed "License Plate Recognition: A Comparative study on Thresholding, OCR and Machine Learning Approaches" using Viola Jones algorithm which consist of four components leading to 80 percent of accuracy.

Kaili Ni [2018] proposed "A Proposed License Plate Classification Method Based on Convolutional Neural Network" which works on basis of seven layers where four are convolutional layer and other three are Max-pooling layer which has accuracy of 98.79 percent.

Table 2.1: Study of Previous Works and Their Accuracy

Author	Year	Title	Accuracy in percent
Saqib Rasheed	2012	Automated number plate recognition using though lines and template matching.	90.62
Alice N Cheeran	2017	Automatic Car Number Plate Recognition.	94

Sumanta Subhadhira	2014	License Plate Recognition Application using Extreme Learning Machines.	89.05
Yoshihiro Shima	2016	Extraction of Number Plate Images Based on Image Category Classification using Deep Learning.	89.7
Wei-chen Liu	2017	A Hierarchical Plate Recognition System using Supervised K-means and Support Vector Machine.	98.89
Yongsheng Li	2019	Vehicle License Plate Recognition Combining MSER and Support Vector Machine in a Complex Environment.	<90
Yang Guang	2017	License Plate Character Recognition Based on Wavelet Kernel LS-SVM.	98.3
Dinesh Bhardwaj	2014	Comparison of ML Algorithms for Identification of Automated Number Plate Recognition.	99
Mahima Satsangi	2018	License Plate Recognition: A Comparative study on Thresholding, OCR and Machine Learning Approaches.	80
Kaili Ni	2018	A Proposed License Plate Classification Method Based on Convolutional Neural Network.	98.79

Motivation:

The motivation behind this project is to enhance security and monitoring measures in buildings, universities, and other premises. By implementing an ANPR system, we aim to automate the vehicle tracking process, enabling seamless identification of vehicles entering and leaving the area. This will help improve security and surveillance and provide valuable data for forensic analysis in case of any security incidents. Additionally, the system will significantly reduce the workload of security personnel, allowing them to focus on more critical tasks and respond promptly to any suspicious activities.

Contribution:

The contribution of automatic number plate detection system is mentioned below. The key contributions of this project will include:

- Development of an ANPR system for tracking vehicles in and out of specific premises.
- Implementation of the YOLO object detection model for real – time number plate localization.
- Integration of Optical Character Recognition (OCR) techniques for accurate character recognition in the segmented number plates.
- Design and implementation of an excel document or Database system to store vehicle movement logs and suspicious activity records.
- Evaluation of the system's performance using real - world camera feeds and comparison with existing ANPR approaches.
- Documentation of the project, including the codebase, data collection, and evaluation results.

III. METHODOLOGY

Problem Formulation:

It is seen that the authorities face problems whenever security forces chase a vehicle or they can't catch a vehicle which has violated traffic rules. Authorities find it very hectic on a busy day to log the vehicle numbers manually. In order to make the entire process of manual data collection to automation we install ANPR system to automatically detect the vehicle which violates the traffic rules, and stores the data of the violating vehicle. The ANPR system can also be used for the vehicle document validation which helps the Authority to check the documents of the vehicle without stopping the vehicle which reduces the human efforts of checking the vehicle document manually one by one. This technology reduces the unnecessary hectic manual work required on any busy day, saves the labour cost. This project is so versatile that it can be used as an entire application once converted to a software or can be used as a part of any major project.

Project Objective:

The project objectives of an Automatic Number Plate Recognition (ANPR) system include accurately detecting and recognizing license plates, enabling vehicle tracking, enhancing security and surveillance, streamlining toll collection and parking management, improving traffic analysis and urban planning, automating law enforcement, providing convenience, and integrating with other systems to adapt to different regions.

Block Diagram of ANPR System:

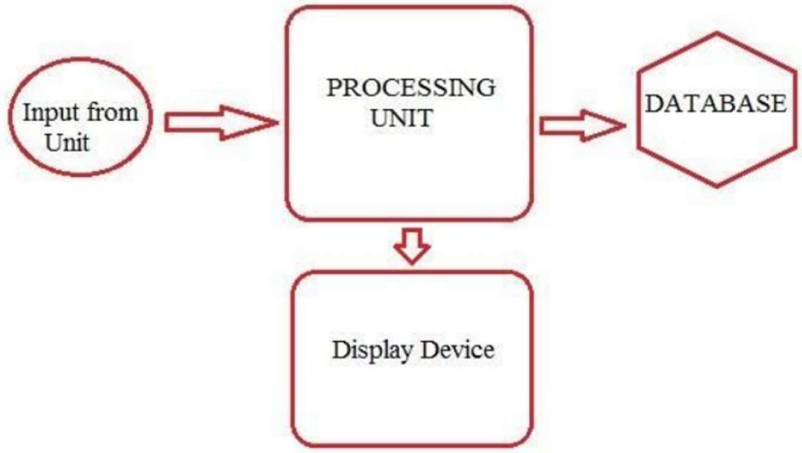


Figure 3.1: Block Diagram of the ANPR System [7]

Methods Used:

ANPR methodology involves capturing images, pre-processing for noise reduction and edge enhancement, detecting license plate regions, segmenting characters, applying OCR techniques for character recognition, validating characters based on context, and assembling the recognized characters into the final license plate number. This process requires a combination of edge detection, contour analysis, character segmentation, recognition algorithms, and validation rules to achieve accurate and efficient results.

Algorithms:

Automatic Number Plate Recognition (ANPR) system uses various algorithm and techniques to improve and achieve its accuracy. Some of the algorithms used in ANPR are as follows:

Thresholding: Thresholding techniques convert grayscale images to binary images, which helps isolate objects of interest.

Hough Transform: The Hough Transform is used for detecting lines and curves in an image. It's often used to identify the straight lines that form the edges of a license plate.

Canny Edge Detection: The Canny edge detector is used to identify edges in the image, which helps in locating object boundaries, including license plates.

K-means: K-means can be used in ANPR for color clustering to handle lighting variations, character feature extraction, and adaptive thresholding. It enhances image quality, aids in character segmentation, and contributes to more accurate recognition.

Neural Networks (CNNs/RNNs): Modern ANPR systems leverage convolutional neural networks (CNNs) or recurrent neural networks (RNNs) for character recognition. These networks learn complex patterns from large datasets.

STEPS OF ANPR MODEL:

To perform an ANPR on a vehicle it needs to follow few steps or stages so that output is obtained the stages can be broadly categorized into four stages they are as follows.

1. Detection and capturing a vehicle
2. Detecting and extraction of number plate
3. Image segmentation technique
4. Performing OCR

The first step is capturing image of the vehicle using the USB camera connected to the PC. The captured image is in RGB format which is converted to greyscale after which different algorithm are performed for the cleaning and extraction of the number plate. Once the number plate is detected the number plate is segmented into each character to perform the OCR and for the data to be extracted.

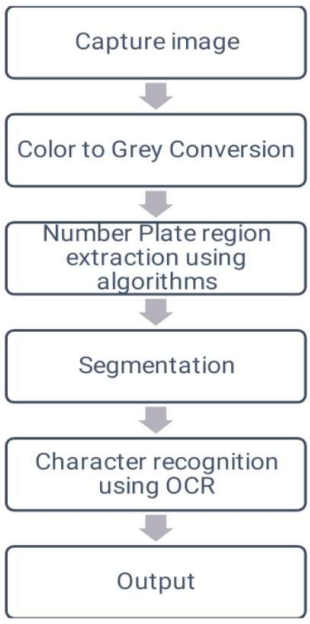


Figure 3.2: Steps of automatic number plate recognition (ANPR) model

Detail of the Process:

The image of a vehicle whose number plate is to be recognised is taken from a digital camera which is then loaded to a local computer for further processing. OpenCV (Open Source Computer Vision) is a library of programming functions mainly aimed at real-time computer vision. To enhance the number plate recognition further we use edge detection further which helps to detect the rectangular edges of the number plate. The ANPR involves further steps:

1. **Capture Image:** This is the first step or process of digital image processing. Image Acquisition is the capturing of an image by any physical device (in this case the primary camera of the computer) so as to take the input as a digital image in the computer. OpenCV is a library used for image processing.

2. **Color to Grey Conversion:** In this step the image is processed to gray scale the data size of the original plate image is the larger one the calculated amount of direct processing these colors are extremely large so the image is converted into gray sale which is convenient for the computer to process the gray scale image.

3. **Number Plate Region Extraction Using Algorithms:** After detection of the vehicle the number plate is detected using Region of Interest. Many algorithms are used for the extraction of number plate, edge detection, k-Means and SVM are some of the algorithms used.

4. **Segmentation:** Segmentation procedures partition an image into its constituent parts or objects. Segmentation is the difficult task in the digital image processing. Segmentation is process of dividing is character into different segment character.

5. **Character Recognition Using OCR:** Optical character recognition or optical character reader is the conversion if images of handwritten or printed text into machine text. OCR is used for the extraction of registered number from the number plate.

6. **Output:** The extracted data is stored in CSV file where the details are further used for the vehicle document validation.

Advantages:

Using ANPR system by automating the car document validation process, we benefit from increased accuracy, time savings, and reduction in fraud and associate costs.

1. **Reduced Human Error:** The presence of ANPR aids for the reduction of manual human error which doesn't fails degree of accuracy.

2. **Efficiency & Time Savings:** ANPR technology allows for almost instantaneous validation, enabling the rapid and high-volume processing of vehicles with an ease.

3. **Cost Reduction:** It reduces the associated costs, and ensures greater compliance.

4. **24/7 System:** As ANPR systems can operate 24 hours a day, they reduce or eliminate the need for security and manned guarding. Moreover it eases overall management.

5. **Tracking of a Vehicle:** ANPR can track vehicle which has violated rules. The presence of chain ANPR can easily notify the last seen of the vehicle.

6. **Reduce Traffic:** It also helps in the reduction of traffic by addition of other algorithm which detects the traffic intensity and controls the traffic flow.

7. **Smart Parking Management:** The present of ANPR can be used for the smart parking management where the Vehicle parking can be charged according to their usage.

8. **Detection of Unauthorized Vehicles:** Vehicles without permits can easily be detected and the unauthorized vehicles can be stopped easily.

Disadvantages:

While automatic number plate detection using computer vision has above advantages, there are also some potential disadvantages and challenges to consider:

1. **Poor Image Resolution:** ANPR performance may degrade when the number plate is too far away from the camera or when using the low- quality cameras, resulting in reduced image resolution and difficulty in accurate plate detection.

2. **Blurry Images:** Motion blur or camera shake can lead to blurred number plate images, making it challenging for the OCR system to accurately recognize characters.

3. **Poor Lighting:** Overexposure, reflections, or shadows in the images can affect the contrast of the number plate, making character recognition difficult.

4. **Obstructed Plates:** License plates may be partially or entirely obstructed by objects like tow bars, dirt, or stickers, which can hinder accurate detection and recognition.

5. **Changing Camera Angle:** If the vehicle changes its lane or angle within the camera's view, it can affect the perspective of the number plate, making it challenging to detect and recognize the plate accurately.

6. **Custom Fonts:** Some vehicles have personalized plates with custom fonts, which can be different from the standard number plate fonts. This adds complexity to character recognition.

7. **Weather Conditions:** Extreme weather conditions such as snow, rain can affect the accuracy and detection of the model.

IV. MODULES

Module -1: Detection and Capturing of a Vehicle

The first stage of ANPR is to detect and capture vehicle. The vehicles are recognised by using YOLO You Only Look Once an algorithm which is used for the detection of object. To detect an object YOLO follows three steps [5]. The first step is dividing into S x S grids the second step is generation of the class probability map the third phase is successfully detection of the object in the frame.

YOLOV3 sets 9 initial anchor box which obtains 9 cluster results (10x13), (16x30), (33x61),(62x45), (59x119), (116x90), (156x198), (373x326). YOLO uses computer vision for the identification and localization of objects within an image or a video. YOLO processes the entire image in a single pass and predicts bounding boxes and class probabilities for various objects. This approach makes YOLO highly efficient and suitable for applications requiring fast and accurate object detection, such as autonomous vehicles, surveillance, and robotics. YOLO's key innovation lies in its ability to perform real-time object detection in a single pass which makes it incredibly fast and accurate compared to other tradition methods.

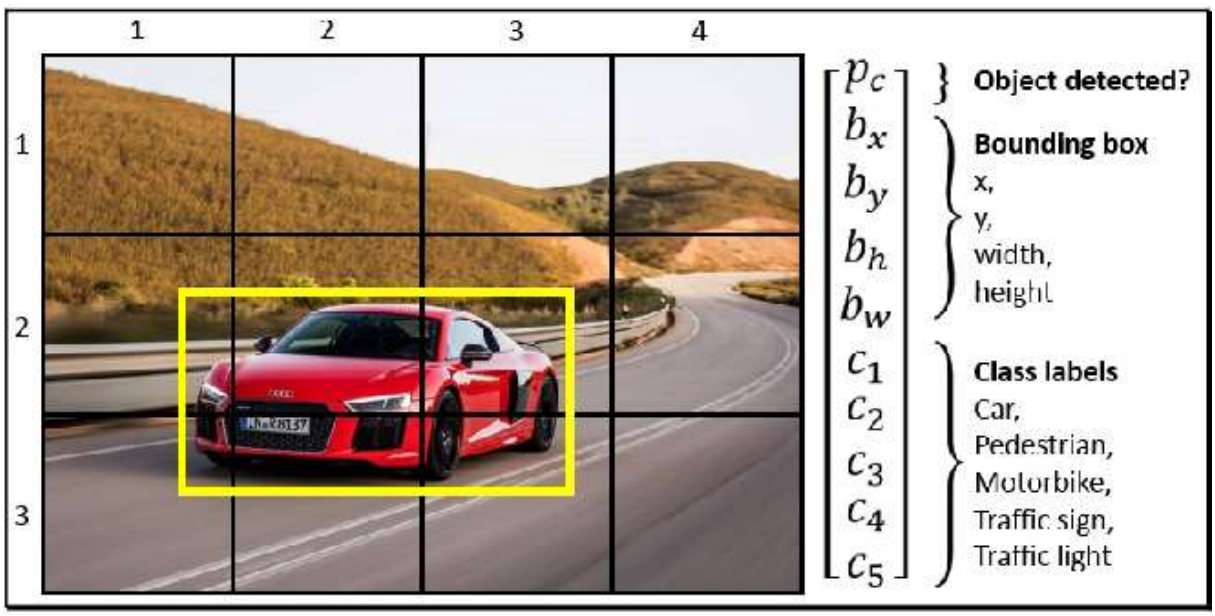


Figure 4.1.1: Bounding box prediction [6]

YOLO algorithm aims to predict a class of an object and the bounding box that defines the object location on the input image. It recognizes each bounding box using four numbers:

- Centre of the bounding box ((b_x, b_y))
- Width of the box (b_w)

- Height of the box (b_h)

In addition to that, YOLO predicts the corresponding number c for the predicted class as well as the probability of the prediction (P_c).

Figure 4.1.2: Detection of Vehicles Using YOLO Algorithm

The COCO (Common Objects in Context) model is used for image recognition which is used for real time object detection. The coco model has 80 object category, 91 stuff categories. The coco model has 17 different key points which is used for pose estimation. In the following project we use four object category for evaluating object detection. The categories 2, 3, 5, 7 of COCO dataset is used these are car, motorbike, bus, truck respectively.

After the Vehicle is detected the image is converted into grey scale, this is because grayscale requires minimum pixel which leading to faster processing times. This efficiency is particularly valuable for real-time applications.

Figure 4.1.3: Conversion of colour image to grey scale image

Code For The Detection and Capturing of The vehicle:

```
from ultralytics import YOLO
import cv2

import util
from sort.sort import *
from util import get_car, read_license_plate, write_csv


results = {}

mot_tracker = Sort()

# load models
coco_model = YOLO('yolov8n.pt')
license_plate_detector = YOLO('./models/license_plate_detector.pt')

# load video
cap = cv2.VideoCapture('./sample.mp4')

vehicles = [2, 3, 5, 7]

# read frames
frame_nmr = -1
ret = True
while ret:
```

```
frame_nmr += 1
ret, frame = cap.read()
if ret:
    results[frame_nmr] = {}
    # detect vehicles
    detections = coco_model(frame)[0]
    detections_ = []
    for detection in detections.boxes.data.tolist():
        x1, y1, x2, y2, score, class_id = detection
        if int(class_id) in vehicles:
            detections_.append([x1, y1, x2, y2, score])

    # track vehicles
    track_ids = mot_tracker.update(np.asarray(detections_))
```

Module-2: Detecting and Extraction of Number Plate

Vehicle after detection using COCO model and YOLO algorithm, the image obtained after the grey conversion is in black and white format. After identify the Region of Interest (ROI), image is then filtered using different algorithm or filtering techniques. The ROI uses Edge Detection Algorithm which makes it easy to detect number plate of a vehicle. Apply edge detection algorithms to identify ROI. Common edge detection algorithms include:

- **Sobel Operator:** Computes the gradient magnitude of the image, highlighting edges by emphasizing areas with high intensity changes.
- **Canny Edge Detector:** This multi-stage algorithm detects a wide range of edges and is known for its good noise suppression and accurate localization of edges.
- **Laplacian of Gaussian (LoG):** This approach combines the Gaussian blur with a Laplacian filter to detect edges.

Once the number plate is detected the number plate is converted binary image to inverted binary image where the thresholding is applied thresholding is performed to convert the gradient magnitude image into a binary image where edges are highlighted. The thresholiding is used according to the lighting condition of the image captured.

Figure 4.2.1: Conversion of binary to Inverted Binary Image.

Module-3: Segmentation of Number Plate

The separation of the number plate plays an important role in the ANPR system and is the difficult task in the digital image processing. Segmentation is process of dividing is character into different segment character. The vehicle number plate are separated to individual character. The separated individual characters are then stored in separate variables.

We get each character with a picture of a number as shown in **Fig 4.3.1.** This process is done so that the each character can be easily identified and OCR can be easily performed for the extraction of the number from number plate. The segmentation of the number plate is done for the improvement of accuracy.

Figure 4.3.1: Segmentation of number plate

After the number plate segmentation is done each segment is passed to Optical Character Reader or Optical Character Recognition (OCR) to segment all the characters in the picture in the form to Text.

Module-4: Performing OCR

Optical character recognition or optical character reader is the conversion if images of handwritten or printed text into machine text. OCR is used for the extraction of registered number from the number plate. ANPR systems use OCR to extract the alphanumeric characters from the detected license plate and convert them into machine-readable text. The **Fig 4.4.1** shows the extraction of the machine-text using OCR. The core of OCR involves recognizing the characters based on their extracted features. There are several OCR algorithms and techniques available, including traditional methods like template matching, as well as more advanced techniques like neural networks.

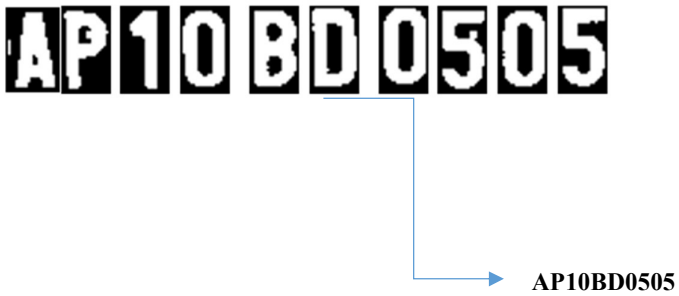


Figure 4.4.1: Optical Character Recognition used after segmentation

After the OCR is performed the machine-text is stored in database for further task to be performed. In this case the machine text is stored in csv format or excel format which is later used to check the document of the vehicle. The vehicle which doesn't have proper document or expired document is store in other database or csv file.

Code For Converting Code To Machine Text:

```
        # detect license plates
        license_plates = license_plate_detector(frame)[0]
        for license_plate in license_plates.boxes.data.tolist():
            x1, y1, x2, y2, score, class_id = license_plate

            # assign license plate to car
            xcar1, ycar1, xcar2, ycar2, car_id = get_car(license_plate, track_ids)

            if car_id != -1:

                # crop license plate
                license_plate_crop = frame[int(y1):int(y2), int(x1): int(x2), :]

                # process license plate
                license_plate_crop_gray = cv2.cvtColor(license_plate_crop, cv2.COLOR_BGR2GRAY)
                _, license_plate_crop_thresh = cv2.threshold(license_plate_crop_gray, 64, 255, cv2.THRESH_BINARY_INV)

                # read license plate number
                license_plate_text, license_plate_text_score = read_license_plate(license_plate_crop_thresh)

                if license_plate_text is not None:
                    results[frame_nmr][car_id] = {'car': {'bbox': [xcar1, ycar1, xcar2, ycar2]},
                                                  'license_plate': {'bbox': [x1, y1, x2, y2],
                                                                    'text': license_plate_text,
                                                                    'bbox_score': score,
                                                                    'text_score': license_plate_text_score}}

# write results
write_csv(results, 'stest.csv')
```

V. RESULTS AND DISCUSSIONS

After performing the ANPR the data is stored in the form of csv format with the number plate extracted. The number plate extracted have repetitive value due to repetition of vehicles in different frames captured. The image **Fig 5.1** shows the extracted data from the video source. The column F shows the extracted machine-text from the number plate detected.

A	B	C	D	E	F	G
frame_nm	car_id	car_bbox	license_plate_bbox	license_plate_bbox_score	license_number	license_number_score
1	5	[2197.398	[2407.91455078125 1	0.595081329	HU51KSU	0.083726241
9	3	[737.1985	[939.8256225585938	0.386960626	NA13NPU	0.257833452
10	5	[2188.893	[2409.128173828125	0.548941672	NI51KSU	0.195882241
14	3	[725.9480	[938.2769165039062	0.563962042	HA13NRU	0.115812475
24	5	[2194.225	[2424.47265625 1619.	0.583319843	NY51KSU	0.094548059
25	5	[2194.873	[2424.463134765625	0.580343068	NY51VSU	0.221632524
28	3	[690.1605	[917.0337524414062	0.446639419	JN41JNR	0.147004185
30	3	[685.1655	[923.3988647460938	0.568454683	NA13NRU	0.254712158
31	3	[683.8938	[923.356201171875 1	0.567542374	NA13NRU	0.254712158
34	5	[2201.105	[2431.63330078125 1	0.651292801	HU51VSV	0.066482291
35	3	[674.2249	[912.9990234375 189	0.577380121	NA13NRU	0.353394622
37	3	[670.8262	[903.099609375 1902.	0.560267568	NA13NRU	0.633190331
39	5	[2203.704	[2432.939453125 166	0.595293045	NI51VSU	0.156837193
40	3	[668.4866	[900.3173217773438	0.619913518	NA13NRU	0.294965051
44	5	[2207.081	[2434.406494140625	0.666397512	NI51VSU	0.226603995
44	3	[656.5110	[877.4298706054688	0.637619555	NA13NRU	0.19947267
45	5	[2206.398	[2440.111328125 168	0.668000221	NV51VSV	0.312712713

Figure 5.1: Output after running program

The repeated number are removed leaving unique number which are detected by ANPR from the video source. The image **Fig 5.2** shows the unique values extracted from the previous CSV file.

Non-Repeated Data			
license_number			
HU51KSU			
NI51KSU			
HA13NRU			
NY51KSU			
NY51VSU			
JN41JNR			
HU51VSV			
NU51ISU			
JH43HPU			
MU51FSU			

Figure 5.2: Extraction of non-repetitive number.

The data in order to check required database for document validation. The database shown here is the government database whose data automatically updates. To represent the government database we have taken a sample of information for a limited vehicles which acts as database. The following image **Fig 5.3** shows all the information about a vehicle which government can have. Which include vehicle number, black list, vehicle colour, fitness of the vehicle, insurance validity date, insurance company, maker of the vehicle, RC Status of the vehicle, Registration date, year of purchase, type of car commercial or personal, pollution certificate validity.

The following data is a dummy database which represent governments database.

Vehicle Number	Blacklist Flag	No. of Fines	Color	Fitness Upto	Insurance Upto	urance Co	Maker	Rc Status	Registration Date	RTO	Year of Purchase	Type of Car	Pollution Cert Validity
NI51KSU	yes	10	Green	2029-06-16	2025-09-16	Oriental In	HERO MO	ACTIVE	2015-12-17	AS	2005	personal	2027-10-18
HA13NRU	yes	4	Black	2030-02-26	2023-09-04	SBI Genera	Honda	ACTIVE	2013-02-07	CN	2003	personal	2023-08-13
NY51KSU	yes	6	Silver	2027-02-12	2028-04-27	SBI Genera	Maruti Suz	DEACTIVE	2010-01-20	HF	2012	commercial	2031-03-18
NY51VSU	no	0	Blue	2031-01-20	2024-07-23	SBI Genera	Honda	ACTIVE	2021-05-02	RC	2018	personal	2025-09-02
JN41JNR	no	0	Green	2029-04-19	2029-02-26	Future Ger	Tata Moto	ACTIVE	2022-09-05	WS	2008	personal	2024-09-19
HU51VSV	no	0	Yellow	2031-11-12	2026-01-13	Bajaj Allian	Maruti Suz	ACTIVE	2016-06-10	UH	2005	commercial	2026-11-26
NU51ISU	yes	5	Green	2026-01-25	2032-12-21	Tata AIG	Hyundai	ACTIVE	2021-02-17	AJ	2014	personal	2027-01-07
JH43HPU	no	0	Silver	2030-12-10	2027-07-11	New India	Maruti Suz	DEACTIVE	2020-11-06	UT	2009	personal	2025-10-23
MU51FSU	yes	6	Green	2028-02-26	2033-06-28	ICICI LOM	Honda	ACTIVE	2014-09-30	DI	2023	commercial	2029-12-02
JN43NRU	no	0	Black	2026-01-15	2025-10-03	ICICI LOM	Toyota	ACTIVE	2023-04-09	FE	2013	personal	2032-11-13
NU51RSU	yes	2	Yellow	2025-06-16	2028-12-24	Future Ger	HERO MO	DEACTIVE	2016-01-07	RM	2004	commercial	2025-02-05
KU51DSU	yes	3	Black	2033-07-27	2023-08-20	New India	Hyundai	ACTIVE	2016-08-29	ZY	2012	commercial	2025-03-17
KU51KSV	no	0	White	2024-01-04	2024-08-18	Oriental In	Honda	ACTIVE	2023-06-12	NS	2004	personal	2032-02-08
MI51GSV	yes	8	Blue	2027-06-18	2027-05-25	United Ind	Hyundai	ACTIVE	2019-03-29	ZH	2015	personal	2028-01-25
NU51ESV	yes	4	Green	2032-04-13	2030-03-30	Bajaj Allian	HERO MO	ACTIVE	2011-02-19	CK	2005	personal	2029-12-08
GX15OCA	no	0	Black	2026-05-15	2024-12-04	New India	HERO MO	ACTIVE	2015-06-30	UQ	2017	personal	2029-09-15
AF05JED	yes	6	White	2030-02-16	2026-04-03	Bajaj Allian	Maruti Suz	DEACTIVE	2014-01-07	IO	2005	personal	2031-12-03
KH05ZZY	no	0	Green	2028-01-04	2027-11-23	Oriental In	HERO MO	ACTIVE	2018-06-15	GT	2016	commercial	2024-11-15
YH05ZIK	no	0	Red	2026-03-21	2025-11-09	Oriental In	Honda	ACTIVE	2016-07-19	JA	2021	commercial	2032-05-11
AF05JEQ	no	0	Silver	2032-10-22	2027-07-03	Tata AIG	Toyota	DEACTIVE	2021-08-20	QZ	2006	commercial	2031-04-01
KW05ZIK	no	0	Red	2033-04-02	2031-10-19	Reliance G	Tata Moto	DEACTIVE	2012-01-09	DF	2010	personal	2032-12-27
AP05JEI	yes	8	Blue	2033-05-23	2030-08-22	HDFC ERG	Tata Moto	DEACTIVE	2017-06-20	LA	2022	commercial	2029-04-14
RN05ZZK	yes	7	Green	2023-11-04	2029-09-09	Future Ger	Toyota	ACTIVE	2021-06-06	LC	2009	commercial	2031-12-24
KH05ZIH	no	0	White	2030-05-01	2029-10-31	SBI Genera	HERO MO	DEACTIVE	2018-06-04	BV	2010	commercial	2026-04-14
RH05ZZI	no	0	Red	2030-04-25	2026-12-22	Reliance G	Maruti Suz	ACTIVE	2019-02-22	KH	2016	commercial	2030-04-29
AP15JEQ	yes	7	White	2031-03-28	2032-11-10	SBI Genera	Tata Moto	DEACTIVE	2021-10-11	JL	2010	personal	2026-06-09
KH65ZZK	yes	7	Silver	2032-07-10	2028-06-08	HDFC ERG	Maruti Suz	DEACTIVE	2018-10-09	PS	2005	commercial	2029-09-12

Figure 5.3: Representation of Government database

The non-repeated data i.e. **Fig 5.1** is compared with government data i.e. **Fig 5.3**. Only the non-repeated number details are taken form the government database and are stored into another csv file this file contains every details of the vehicle present in the video. The following image shows the extraction of data from government database **Fig 5.4**.

	NI51KSU	yes	10	Green	16-06-2029	16-09-2025	Oriental In	HERO MO	ACTIVE	17-12-2015	AS	2005	personal	18-10-2027
	HA13NRU	yes	4	Black	26-02-2030	04-09-2023	SBI Genera	Honda	ACTIVE	07-02-2013	CN	2003	personal	13-08-2023
	NY51KSU	yes	6	Silver	12-02-2027	27-04-2028	SBI Genera	Maruti Suz	DEACTIVE	20-01-2010	HF	2012	commercial	18-03-2031
	NY51VSU	no	0	Blue	20-01-2031	23-07-2024	SBI Genera	Honda	ACTIVE	02-05-2021	RC	2018	personal	02-09-2025
	JN41JNR	no	0	Green	19-04-2029	26-02-2029	Future Ger	Tata Moto	ACTIVE	05-09-2022	WS	2008	personal	19-09-2024
	HU51VSV	no	0	Yellow	12-11-2031	13-01-2026	Bajaj Alliar	Maruti Suz	ACTIVE	10-06-2016	UH	2005	commercial	26-11-2026
	NU51ISU	yes	5	Green	25-01-2026	21-12-2032	Tata AIG	Hyundai	ACTIVE	17-02-2021	AJ	2014	personal	07-01-2027
	JH43HPU	no	0	Silver	10-12-2030	11-07-2027	New India	Maruti Suz	DEACTIVE	06-11-2020	UT	2009	personal	23-10-2025
	MU51FSU	yes	6	Green	26-02-2028	28-06-2033	ICICI LOM	Honda	ACTIVE	30-09-2014	DI	2023	commercial	02-12-2029
0	JN43NRU	no	0	Black	15-01-2026	03-10-2025	ICICI LOM	Toyota	ACTIVE	09-04-2023	FE	2013	personal	13-11-2012
1	NU51RSU	yes	2	Yellow	16-06-2025	24-12-2028	Future Ger	HERO MO	DEACTIVE	07-01-2016	RM	2004	commercial	05-02-2025
2	KU51DSU	yes	3	Black	27-07-2033	20-08-2023	New India	Hyundai	ACTIVE	29-08-2016	ZY	2012	commercial	17-03-2025
3	KU51KSV	no	0	White	04-01-2024	18-08-2024	Oriental In	Honda	ACTIVE	12-06-2023	NS	2004	personal	08-02-2032
4	MI51GSV	yes	8	Blue	18-06-2017	25-05-2027	United Ind	Hyundai	ACTIVE	29-03-2019	ZH	2015	personal	25-01-2028
5	NU51ESV	yes	4	Green	13-04-2032	30-03-2030	Bajaj Alliar	HERO MO	ACTIVE	19-02-2011	CK	2005	personal	08-12-2029
6	GX15OCA	no	0	Black	15-05-2026	04-12-2024	New India	HERO MO	ACTIVE	30-06-2015	UQ	2017	personal	15-09-2029
7	AF05JED	yes	6	White	16-02-2010	03-04-2026	Bajaj Alliar	Maruti Suz	DEACTIVE	07-01-2014	IO	2005	personal	03-12-2011
8	KH05ZZY	no	0	Green	04-01-2028	23-11-2027	Oriental In	HERO MO	ACTIVE	15-06-2018	GT	2016	commercial	15-11-2024
9	YH05ZIK	no	0	Red	21-03-2026	09-11-2025	Oriental In	Honda	ACTIVE	19-07-2016	JA	2021	commercial	11-05-2032
0	AF05JEQ	no	0	Silver	22-10-2032	03-07-2027	Tata AIG	Toyota	DEACTIVE	20-08-2021	QZ	2006	commercial	01-04-2031

Figure 5.4: Document Details of the vehicle extracted using ANPR

To make things much simpler the data is further most processed and only the required information is represented. The program uses current date and time module and checks whether the documents are active or expired if the date present in the csv colum is less than the current date it shows "expired". The active or expired data make it easy for the authorities to easily check the data with an ease. The following image **Fig 5.5** shows the easy representation of vehicles document.

A	B	C	D	E	F
Vehicle Number	Fitness	Insurance	Rc	Pollution Cert Validity	
NI51KSU	Valid	Valid	ACTIVE	Valid	
HA13NRU	Valid	Expired	ACTIVE	Valid	
NY51KSU	Valid	Valid	DEACTIVE	Valid	
NY51VSU	Valid	Valid	ACTIVE	Valid	
JN41JNR	Valid	Valid	ACTIVE	Valid	
HU51VSV	Valid	Valid	ACTIVE	Valid	
NU51ISU	Valid	Valid	ACTIVE	Valid	
JH43HPU	Valid	Valid	DEACTIVE	Valid	
MU51FSU	Valid	Valid	ACTIVE	Valid	

Figure 5.5: Representation of Vehicle document

The total process of document verification can be explained using image which represents the steps by step procedure for document verification. The image below shows the overall process of document verification **Figure 5.6**.

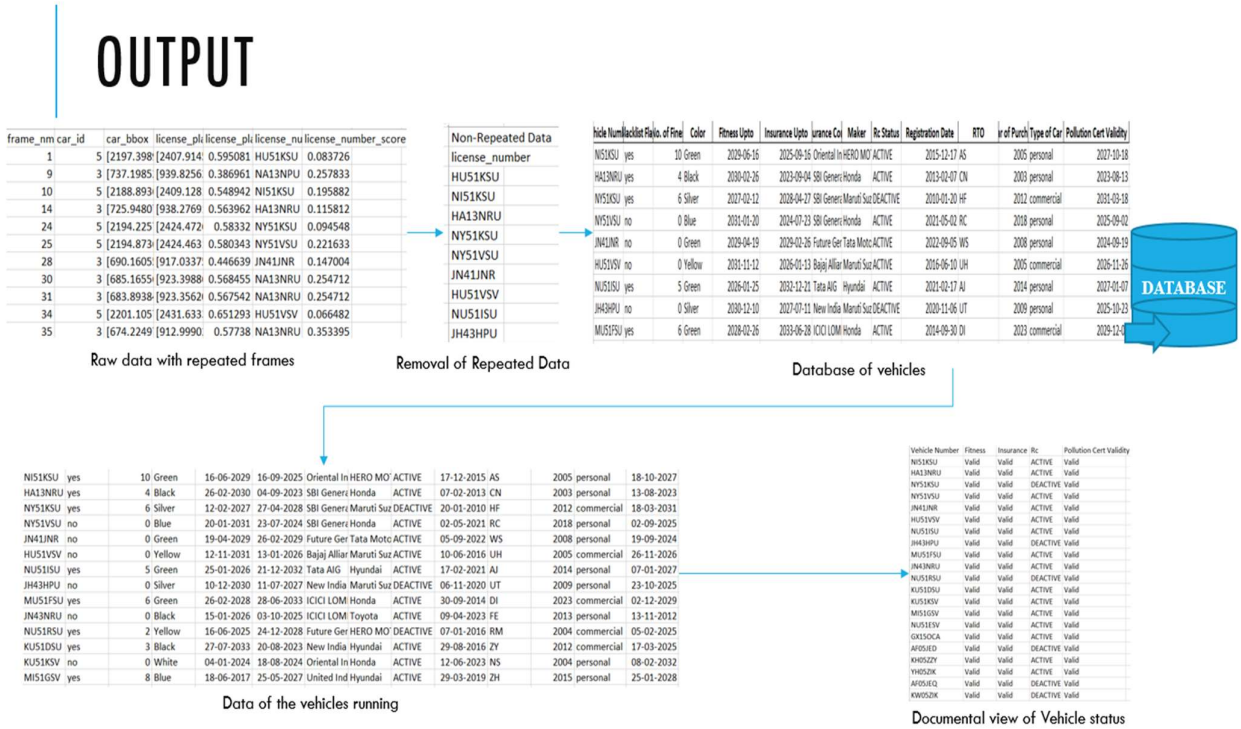
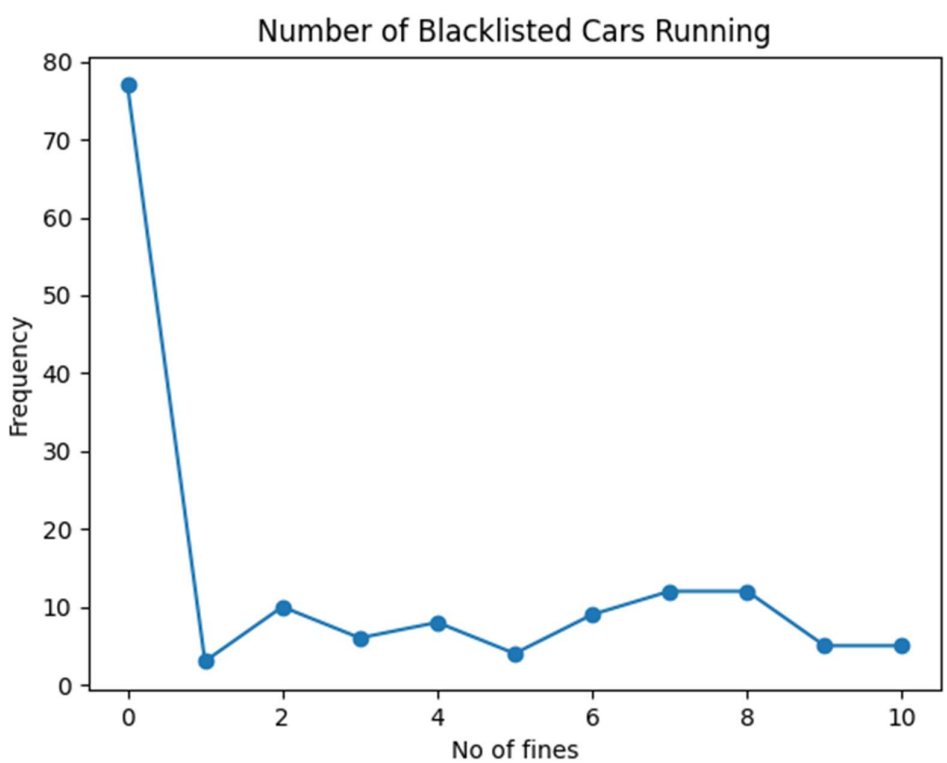


Figure 5.6: Overall steps for document verification

For the better understanding of vehicle document status the authoritative can check the graphical representation, which depicts the graphical image of the overall vehicles passed and their document status.
The graphical representation gives authorities a clear view of documents which average vehicle owners don't renew.

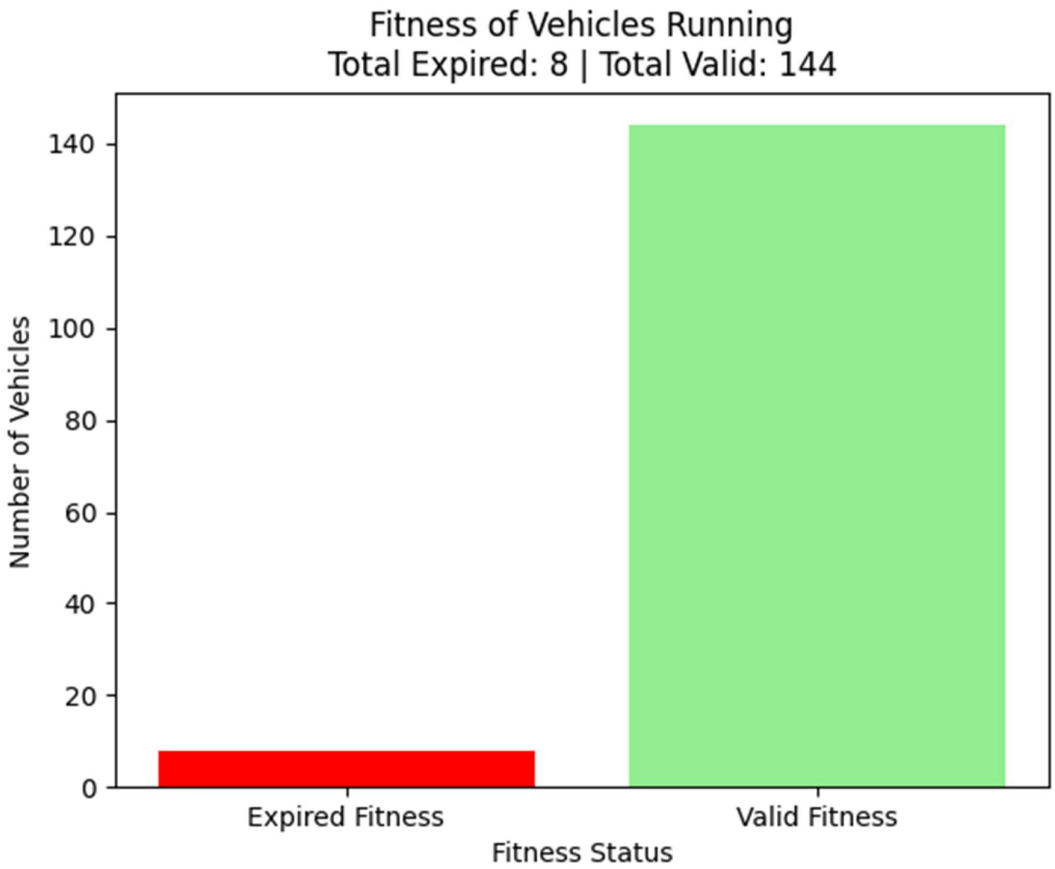
When a Vehicle is caught by cops or other Authorities if the vehicle is subjected to any violation of any rules the vehicle owner is made to pay penalty. The penalty is registered to using the licence number of the number plate. The number of fines on a vehicle represents the whether the Vehicle is blacklisted or not. The vehicle is with zero fines is not blacklisted but the vehicle which has one or greater penalty registered is subjected to blacklisted. To find the overall blacklisted vehicles the graph is used.

The following graph represents the blacklisted vehicles running on the particular day. On the X axis is the representation of the number of fines imposed on a vehicle, on the Y axis the graph shows no of vehicles having number of fines register.



Graph 5.1: Representation of number of blacklisted vehicles running.

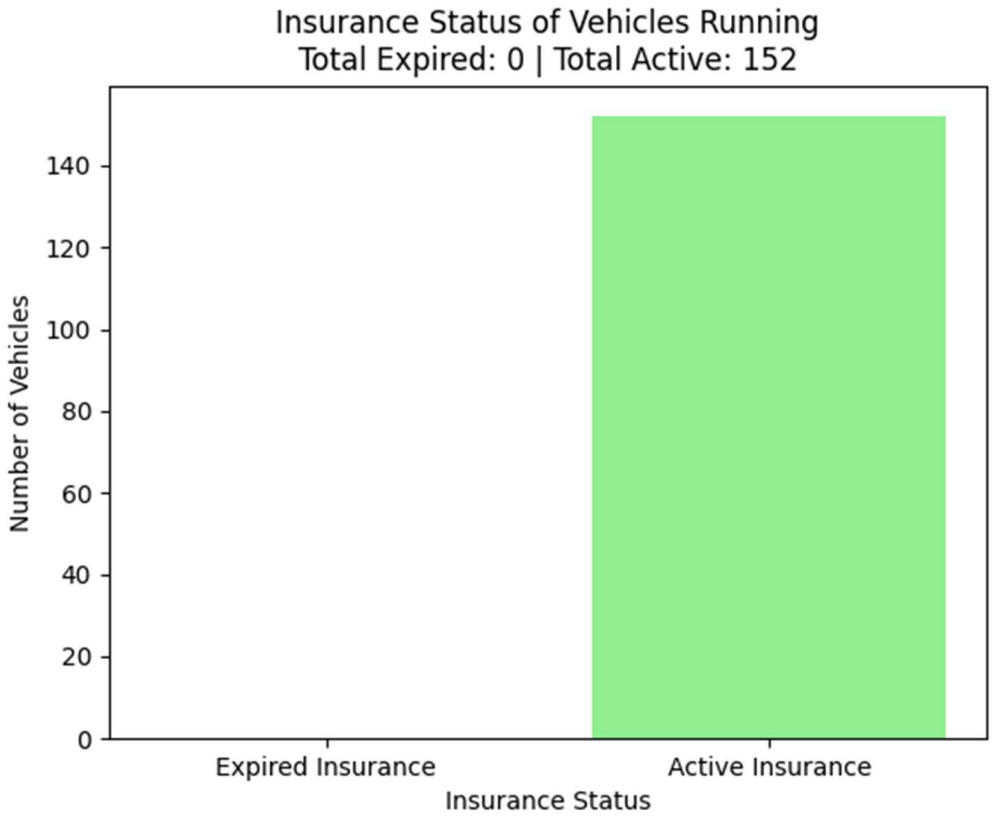
The vehicle fitness depends on the number of years left for a Vehicle to run on road. The vehicle have certain amount of years after which the vehicle is unfit for running on road. If the vehicle continues to run still after the expiry of the fitness of the vehicle it is subjected to penalty.
The following graph represents the Fitness of the vehicle running. On the X axis represents the fitness status “valid” or “expired” on the Y axis it represents the number of vehicles. The graph also shows the total number of expired and total valid vehicle on the top of the graph.



Graph 5.2: Representation Fitness of the vehicle

For a vehicle insurance document is one of the important document these document are the document which are checked by cops. Which makes it important for the vehicle owners to maintain insurance paper and renew it when it is about to expire.

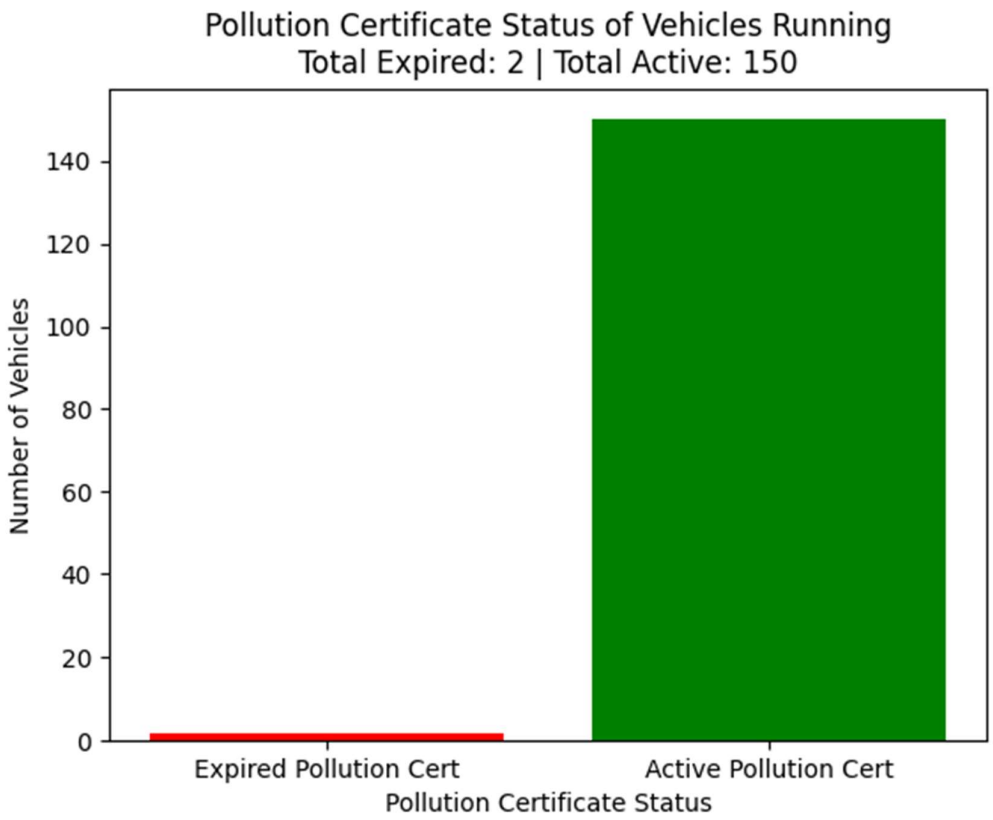
On the X axis represents the insurance status “valid” or “expired” on the Y axis it represents the number of vehicles. The graph also shows the total number of expired insurance and total active insurance active vehicle on the top of the graph. The below graph shows the graphical representation of the insurance status of the vehicle.



Graph 5.3: Representation of vehicle insurance status

A PUC (Pollution under Control) certificate is used to indicate that a vehicle's emissions are within the acceptable limits set by the environmental authorities. If a Pollution under Control (PUC) certificate for a vehicle has expired, it usually implies that the vehicle hasn't undergone the required emission testing within the specified period. The consequences of driving a vehicle with an expired PUC certificate can lead to legal penalty.

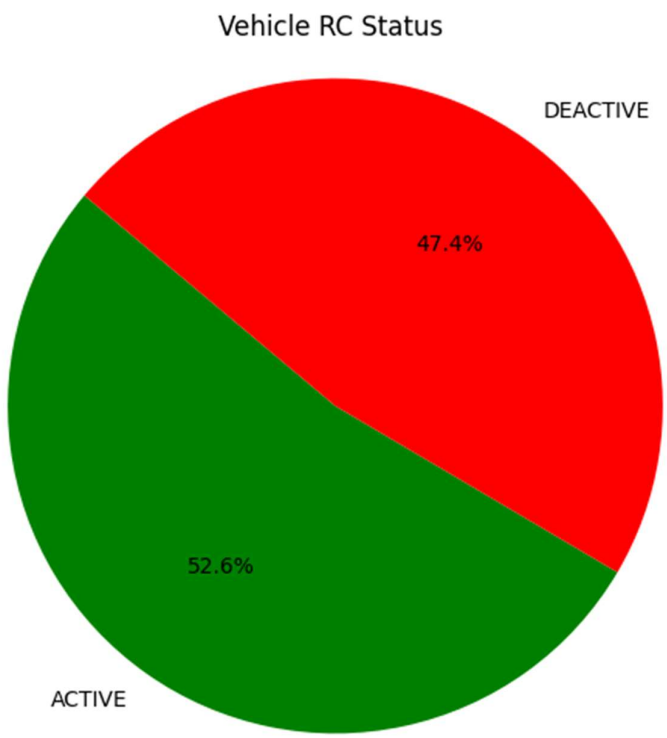
On the X axis represents the PUC status "valid " or "expired " on the Y axis it represents the number of vehicles. The graph also shows the total number of expired PUC and Total active PUC active.



Graph 5.4: Representation of vehicle PUC status.

The term "RC of a vehicle" likely refers to the "Registration Certificate" of a vehicle. The Registration Certificate is an official document issued by the government's vehicle registration authority that serves as proof of the vehicle's registration. It contains important information about the vehicle and its owner. An RC (Registration Certificate) can become inactive when a vehicle is permanently taken out of service, ownership is transferred, the vehicle is exported, there's non-renewal of registration, the vehicle reaches its maximum life, or due to violations of regulations. The exact process varies based on local laws.

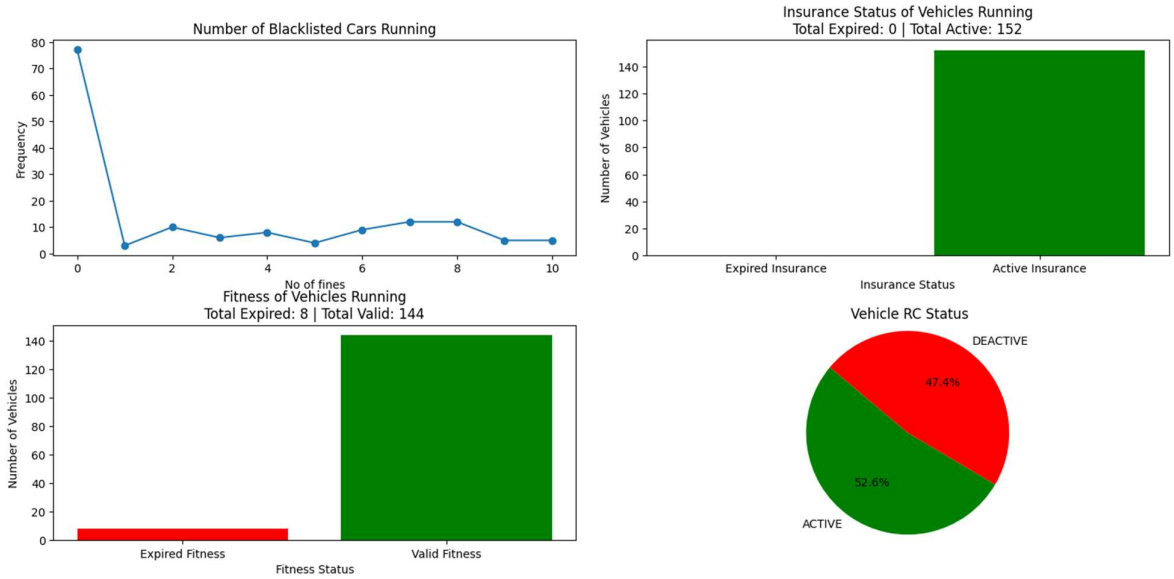
The active and deactive state of RC is represented in the form of pie chat where the green colour represents the active status whereas the red represents the deactivate status.



Graph 5.5: Pie Representation of RC status

Checking each graph for an authoritative can be a tedious task to make work more easier the graphs can be easily represented into a one format where an authoritative can easily check all

graph all at once without the struggle of checking each and every graph one by one. The following graph shows all resonation of graph all together.



Graph 5.6: Representation of all graph in one graph

VI. CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion: The project is designed in a way that we can understand and implement technology used in day-to-day life. Automatic Number Plate Recognition is mostly used in developed countries like Singapore, Japan, Germany, etc. Implementation of ANPR can reduce human effort.

It is seen that security forces all over the world face problem to locate vehicle which have violated rules. It is also seen that technology can greatly help us in this situation by solving it.

The ANPR is growing technology in the field of machine learning, this technology would help county to maintain proper law and order.

Future Enhancement:

The ANPR can be developed further by increasing its accuracy and merging of other technologies like triple ride detection for completer Road monitoring system.

The system can also help in monitoring of traffic, which helps in traffic control analysis. The ANPR system can be further developed by capturing images and by enabling time and location of the vehicle which have violated traffic rules.

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